

Digital Circuits

Exercise 2 - Basic Arithmetic and Status Flags

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Total points	47	

Part 1 - Digital Adder

P1.1 - Full Adder Cell

Task Description:

The objective is to design a 1-bit Full Adder (FA) cell. A Full Adder is a digital circuit that performs the addition of three inputs: two significant bits (A and B) and an incoming carry (C_{in}). It produces two outputs: the Sum (S) and the Carry Out (C_{out}).

Truth Table

The following table describes the functionality of the 1-bit Full Adder implemented.

Ci (Carry In)	B	A	Co (Carry Out)	S (Sum)
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Table 1: Truth Table for the Full Adder Cell

Solution: Boolean Equations

Based on the truth table above, the Boolean equations for the Sum (S) and Carry Out (C_{out}) are derived as follows:

Sum (S): The sum bit is high when an odd number of inputs are high. This corresponds to the XOR operation.

$$S = A \oplus B \oplus C_{in}$$

Carry Out (C_{out}): The carry out is high when two or more inputs are high. This can be expressed as:

$$C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B))$$

T1.2 - Timing Analysis (Adder)

Lab Procedure

We performed a timing analysis on the synthesized design of the simple 8-bit Ripple Carry Adder. Using the Vivado "Report Timing Summary," we inspected the unconstrained setup paths to determine the propagation delay.

Theory

Logic Synthesis results provide timing estimates. At this stage, the tool knows the logic gates used and their individual delays –the logic delay– from its internal library. However, since the components haven't been placed on the actual FPGA chip, the wire delays –the net delay– are estimated using statistical models.

Physical Synthesis involves the actual rewiring on the FPGA internal structure. The timing analysis performed after implementation is much more accurate because it calculates the net delay based on the actual physical length and electrical properties –resistance and capacitance– of the routed wires.

Results

The timing results for the simple adder are shown in Figure 1. The critical path (Path 1) characteristics are:

Metric	Value (ns)
Total Data Path Delay	8.553
Logic Delay	5.546
Net (Routing) Delay	3.007

Table 2: Timing Analysis Results for Simple Adder (T1.2)

Conclusion

The timing delay table shows the time it takes for a signal to reach the output port. The total number of paths is 9, which corresponds to all possible outcomes / combinations possible.

The physical synthesis, unlike the logic synthesis, shows information gathered from a real FPGA chip that is being programmed. That's why the results may differ depending on the chip condition.

Part 2 - Subtractor Extension and Status Bits

T2.2 - Timing Analysis (Addsub)

Lab Procedure

In this task, the design was modified to include subtraction capabilities using a control input (*sel*) and multiplexers. We ran the timing analysis again to observe the impact of this additional logic on the critical path.

Results

The timing results for the Adder/Subtractor are shown in Figure 3. The critical path characteristics are:

Metric	Value (ns)
Total Data Path Delay	8.851
Logic Delay	5.578
Net (Routing) Delay	3.284

Table 3: Timing Analysis Results for Adder/Subtractor (T2.2)

Conclusion

The delays for the Adder / Subtractor setup are slightly higher than those of a simple adder. This increase is due to the modification of the design, adding the control input (*sel*) and multiplexers increased the complexity of the circuit, inserting more logic gates into the critical path and increasing the overall delay. This can be observed in Figures 2 and 4.

Appendix: Timing Diagrams

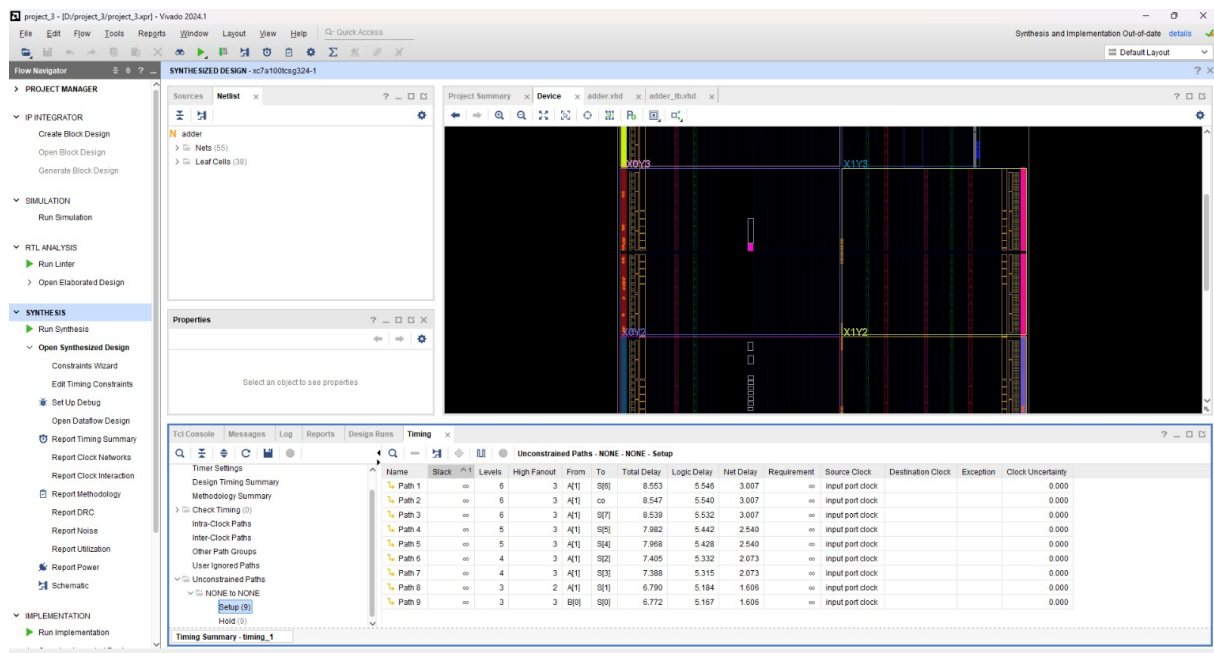


Figure 1: Vivado Timing Summary for the Simple Adder (T1.2). Max Delay: 8.553 ns.

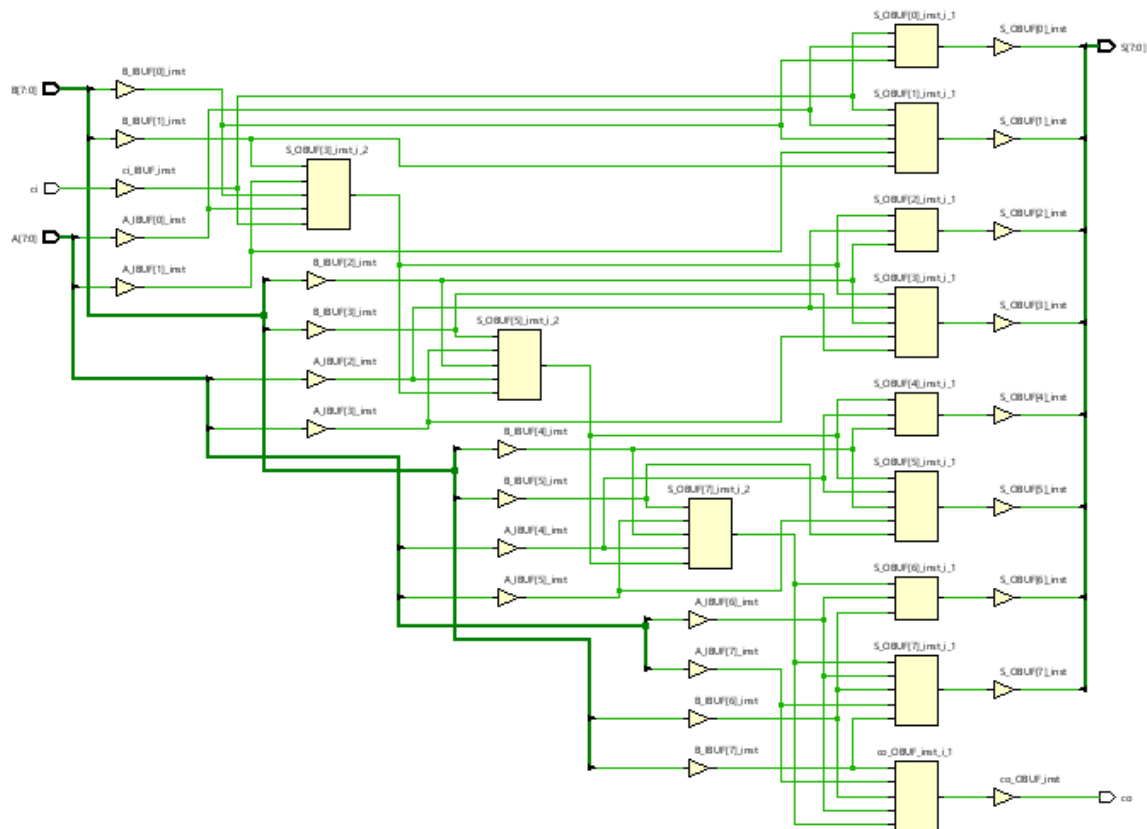


Figure 2: Vivado Timing Summary for the AddSub Module Schematic (T1.2).

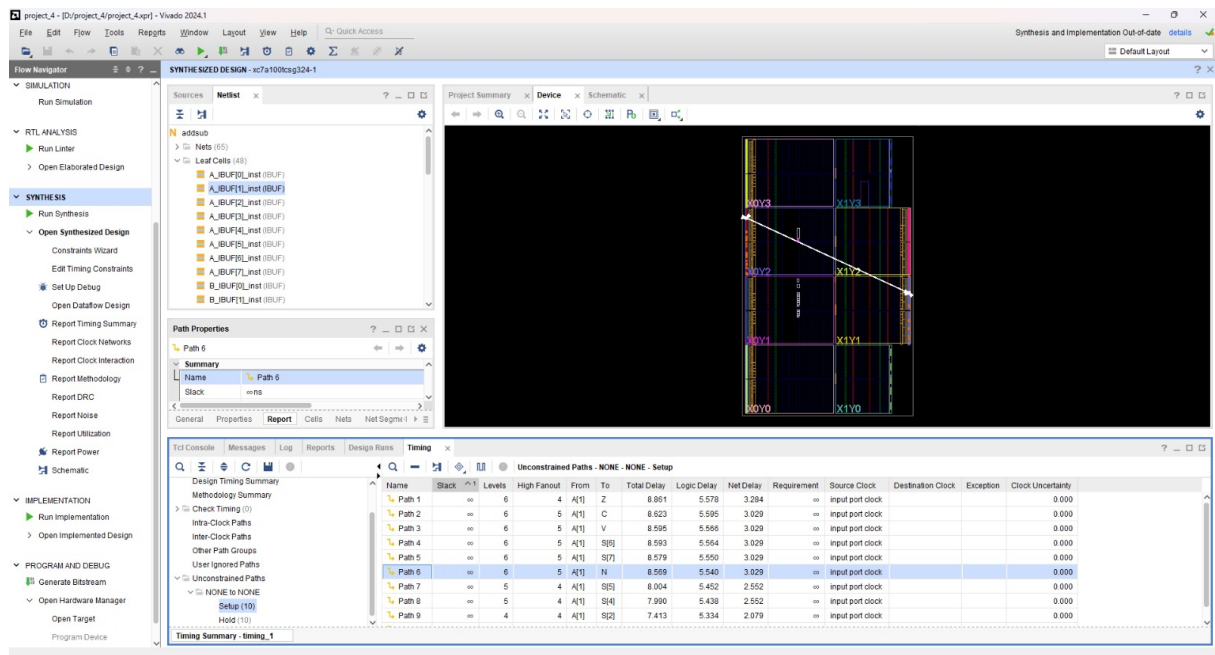


Figure 3: Vivado Timing Summary for the AddSub Module (T2.2). Max Delay: 8.851 ns.

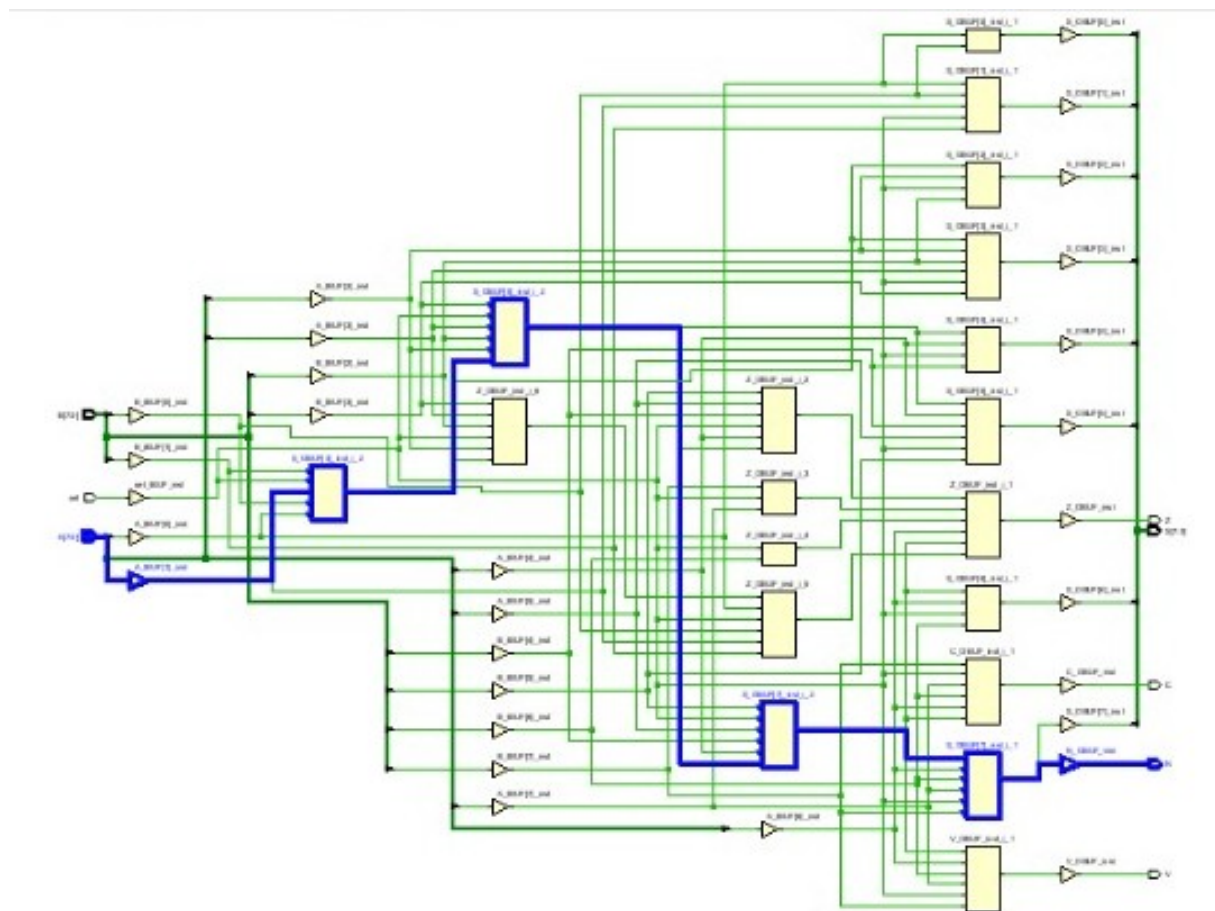


Figure 4: Vivado Timing Summary for the AddSub Module Schematic (T2.2).